Mathilde Perez

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Education

Télécom Paris — *Palaiseau, France*
PhD in Graph Learning, Fairness in AI — *Sept 2023 – present*
- Fairness constraints on evolutionary graph learning models
- Evaluation framework for biased vs. unbiased link prediction models

Université Paul Sabatier — *Toulouse, France*
Master in Applied Mathematics for Engineering, Industry and Innovation — *2020 – 2022*
- AI, Deep Learning, Machine Learning, Big Data, image processing
- Optimization, Statistics, Stochastic Simulations, PDEs, Signal Processing

Université Paul Sabatier — *Toulouse, France*
Bachelor in Applied Mathematics — *2017 – 2020*

Lycée des Arènes — *Toulouse, France*
Baccalauréat in Sciences — *2016 – 2016*
- Honors (Mention Bien), European English section

Publications

How Predicted Links Influence Network Evolution: Disentangling Choice and Algorithmic Feedback in Dynamic Graphs — 2026
Mathilde Perez, Raphaël Romero, Jefrey Lijffijt, Charlotte Laclau
arxiv.org/abs/2603.03945 (UAI 2026)

TopoFair: Linking Topological Bias to Fairness in Link Prediction Benchmarks — 2026
Lilian Marey, Mathilde Perez, Tiphaine Viard, Charlotte Laclau
arxiv.org/abs/2602.11802 (Preprint)

SimHawNet: a modified Hawkes process for temporal network simulation — 2025
Mathilde Perez, Raphaël Romero, Bo Kang, Tijl De Bie, Jefrey Lijffijt, Charlotte Laclau
10.1007/s10618-025-01119-1 (Data Mining and Knowledge Discovery)

Experience

Research Engineer — *Palaiseau, France*
Télécom Paris — *Apr 2023 – Sept 2023*
- Simulation of temporal graphs with Hawkes processes

Human-Computer Interaction Intern — *Paris, France*
ISIR — Sorbonne / CNRS — *Apr 2022 – Oct 2022*
- Modeling social attention through gaze in a conversational agent (Greta platform)

NLP Intern — *Purpan, France*
CerCo — CNRS — *May 2021 – July 2021*
- Similarities between CLIP and the brain

Oral Presentations

- **SimHawNet: a modified Hawkes process for temporal network simulation** — ECML-PKDD 2025, Porto, Portugal · Slides & CAp 2024, Lille, France
- **How social media algorithms reinforce inequality** — EUNIC Science Day, Madrid, Spain · Slides

Skills

Programming
Python, PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy, Matplotlib, C/C++/C#, R, LaTeX, Unity, Java

Machine Learning & AI
Graph Learning, Deep Learning, Fairness in AI, Temporal Graphs, Hawkes processes

Soft Skills
Teamwork, Public speaking, Autonomy

Languages

French
Native

English
Advanced